

1. Aurora Over an Arctic Landscape

For $m = 1, \dots, 2000$, $n = 1, \dots, 1200$,

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_1\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_2\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

and

$$\begin{aligned} H_v(x,y) = & \frac{21}{20}A_1(x,y)W(x,y)e^{-e^{-10+10x}} \\ & + \left(\frac{10-5v+3v^2}{40}\left(\frac{3}{2}-y\right) + 2R_1(x,y)e^{-e^{-10|y-\frac{3}{2}|^2}}\right)\left(\prod_{s=1}^{50}(1-J_s(x,y))e^{-e^{-10^?}}\right) \\ & + \frac{20-11v+5v^2}{40}(1-W(x,y))e^{-e^{-10xy}} \\ & + \frac{34-7v+5v^2}{40}\left(\frac{3}{5}+\frac{7}{10}e^{-e^{\Phi(x,y)}}\right)\left(5-2e^{-e^{\Psi(x,y)}}\right)e^{-e^{\Omega(x,y)}}, \end{aligned}$$

$$B_v(x,y)=G_v(x,y)+40\left(y+\frac{1001}{10000}\right)e^{-e^{\Theta(x,y)}},$$

$$G_v(x,y)=\sum_{s=1}^{50}\left(\frac{193}{200}\right)^se^{-\Gamma_s(x,y)},$$

$$R_v(x,y)=\sum_{s=1}^{50}\left(\frac{5}{2}+\frac{5}{2}(v-1)^2+(3-3v)(-1)^s\right)e^{-\Lambda_s(x,y)},$$

$$W(x,y)=e^{-e^{20\left(y+\frac{1001}{10000}+\frac{x}{50}\right)}},$$

$$A_v(x,y)=\sum_{s=1}^{20}\left(\prod_{u=0}^{s-1}(1-J_u(x,y))\right)J_s(x,y)\left(\frac{1+v}{2}-\frac{7v^2-7v}{10}+\frac{(-1)^s}{80}+\frac{1}{10}\cos(2Q_s(x,y)+\cos((v+1)Q_s(x,y)))\right),$$

$$V_v(x,y)=\left(y+\frac{1001}{10000}\right)^{-1}\left(1-\frac{x}{4}\right)+\frac{v}{100},$$

$$J_v(x,y)=\frac{47^v}{50^v}e^{-e^{-i\cos(-y+\frac{1}{2})}}e^{-e^{-\frac{v}{7}\cos(\cdots)}}e^{-e^{i\Theta(x,y)}},$$

$$U_v(x,y)=\left(y+\frac{1001}{10000}\right)^{-1}\left(x+\frac{1}{9}\right),$$

$$\begin{aligned} Q_s(x,y) = & 3\left(x+\frac{s}{500}\right)K_s(y)+2\cos(K_s(y))+\frac{2}{5}\cos\Big(5\left(x+\frac{s}{500}\right)K_s(y)+8K_s(y)\Big) \\ & +\frac{1}{10}\cos\Big(15\left(x+\frac{s}{500}\right)K_s(y)-18K_s(y)\Big), \quad K_s(y)=\left(\frac{50}{49}\right)^s\left(y+\frac{1001}{10000}\right)^{-1}. \end{aligned}$$

2. Interacting Black Holes

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{500}, \frac{601-n}{500}\right)\right), F\left(H_1\left(\frac{m-1000}{500}, \frac{601-n}{500}\right)\right), F\left(H_2\left(\frac{m-1000}{500}, \frac{601-n}{500}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$H_v(x,y)=e^{-e^{-50\left(y-\frac{\pi}{3}-\frac{7}{50}\right)}}\frac{B(x,y)}{5}+\frac{20K_v(x,y)}{37}\left(1-e^{-e^{200\left(\left(x+\frac{1}{3}\right)^2+\left(y-\frac{3}{100}\right)^2-\frac{3}{100}\right)}}e^{-200\left(y-\frac{\pi}{3}-\frac{5}{5}\sqrt{\frac{3}{100}\left(\left(x+\frac{1}{3}\right)^2+\left(y-\frac{3}{100}\right)^2}\right)}\right)\right)\\ \times\left(1-e^{-e^{200\left(\left(x+\frac{1}{3}\right)^2+\left(y-\frac{3}{100}\right)^2-\frac{1}{50}\right)}}e^{-200\left(y-\frac{\pi}{3}-\frac{5}{5}\sqrt{\frac{1}{50}\left(\left(x+\frac{1}{3}\right)^2+\left(y-\frac{3}{100}\right)^2}\right)}\right)\right),$$

$$B(x,y)=\sum_{s=1}^{50}e^{-e^{-100\left(\sin^6(T_s(x,y))-\frac{199}{200}\right)}}-e^{-\frac{\cos^6(200T_s(x,y)+s^2)}{20}}-e^{1000\left(|x+\frac{y}{3}|-\frac{7}{10}\right)},$$

$$T_s(x,y)=\pi\frac{3x+y}{3}+\frac{7+5\cos(s^2)}{35}\left(y-\frac{x^2}{3}\right)\cos\left((6+3\cos(4s^2))\frac{3y-x}{3}+2\cos(4y+2x+5s^2)+3s^2\right),$$

$$K_v(x,y)=\sum_{s=1}^{50}\left(\frac{19}{20}\right)^s\left(\frac{3}{2}+\frac{1}{2}(v-1)^2+\frac{4-4v}{10}\cos(s^2)\right)\\ \times e^{-e^{-\frac{3}{2}\left(\cos\left(\left(\frac{3}{4}\right)^sL_{1,s}(x,y)+\cos\left(\left(\frac{3}{4}\right)^sL_{3,s}(x,y)\right)+2\cos(32s^2)\right)\left(\frac{3}{4}\right)^s\left(\sin(s^2)U_v(x,y)-\frac{3}{2}\cos(s^2)V_v(x,y)\right)+\cos\left(\left(\frac{3}{4}\right)^sL_{5,s}(x,y)+2\cos(12s^2)\right)-\frac{3}{2}A_s(x,y)\right)}},$$

$$A_s(x,y)=e^{-\frac{1}{2}\sqrt{\left(x+\frac{2}{3}\right)^2+6\left(y-\frac{2}{3}\right)^2}\sqrt{\left(x+\frac{2}{3}\right)^2+6\left(y-\frac{2}{3}\right)^2}U(x,y)+\cos^2(7s^2)-\frac{7}{20}}\\ +e^{10\left(\left(x+\frac{2}{3}\right)^2+6\left(y-\frac{2}{3}\right)^2-\frac{3}{100}\right)}+e^{10\left(\left(x+\frac{2}{3}\right)^2+6\left(y-\frac{2}{3}\right)^2-\frac{3}{100}\right)},$$

$$L_{v,s}(x,y)=\cos(us^2)U(x,y)+\frac{3}{2}\sin(us^2)V(x,y),$$

$$U(x,y)=\cos\big(2\ln\big((Q(x,y))^2+(P(x,y))^2\big)\big)\,Q(x,y)+\sin\big(2\ln\big((Q(x,y))^2+(P(x,y))^2\big)\big)\,P(x,y),$$

$$V(x,y)=\arctan\left(\frac{Q(x,y)}{|P(x,y)|}\right),$$

$$P(x,y)=\frac{3y-x}{3}\left(\frac{3}{2\left(x+\frac{y}{3}+\frac{1}{2}\right)^2+12\left(y-\frac{1}{3}\right)^2+\frac{1}{10^5}}+\frac{1}{\left(x+\frac{y}{3}-\frac{1}{2}\right)^2+6\left(y-\frac{1}{3}\right)^2+\frac{1}{10^5}}\right),$$

$$Q(x,y)=\frac{3x+y+\frac{3}{2}}{2\left(x+\frac{y}{3}+\frac{1}{2}\right)^2+12\left(y-\frac{1}{3}\right)^2+\frac{1}{10^5}}+\frac{x+\frac{y}{3}-\frac{1}{2}}{\left(x+\frac{y}{3}-\frac{1}{2}\right)^2+6\left(y-\frac{1}{3}\right)^2+\frac{1}{10^5}}.$$

3. Swarm of Bats

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{600}, \frac{501-n}{600}\right)\right), F\left(H_1\left(\frac{m-1000}{600}, \frac{501-n}{600}\right)\right), F\left(H_2\left(\frac{m-1000}{600}, \frac{501-n}{600}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$H_v(x,y)=\frac{2-v}{10}(1-A(x,y))\left(1-\frac{4}{5}R(x,y)\right)\\ +\frac{11}{10}A(x,y)\left(B_v(x,y)\left(1-\frac{y}{6}\right)(1-R(x,y))+\left(\frac{3}{25}-\frac{3v^2-3v}{200}+\frac{E(x,y)}{50}\right)R(x,y)\right),$$

$$\begin{aligned}
B_v(x,y) &= \frac{v^2-v+8}{10} + \frac{3}{20}e^{-e^{150\left[\frac{2}{5}\left(\frac{1}{2}+\frac{1}{\pi}\arctan\left(-70\left(x+\frac{y}{2}\right)\right)\right)\right]\left[\left(x+\frac{y}{3}\right)^2+\left(y-\frac{x}{4}\right)^2\right]-15}}, \\
R(x,y) &= e^{-e^{100\left(y+\frac{y}{10}+\frac{x}{16}\sin(4x+\cos(3x))+\frac{E(x,y)}{20}\right)}}, \quad A(x,y) = \prod_{s=1}^{32} U_s(x,y)V_s(x,y), \\
V_s(x,y) &= e^{-e^{-\frac{700}{100}\left(\frac{107}{100}\right)^{-s}\left((2-\cos(ss^2))^{2/5}P_s(x,y)-\frac{7}{100}|\cos(ss^2)|^{2/5}-\frac{1}{100}\sin^4(7Q_s(x,y))-4200\left(\frac{107}{100}\right)^{-s}Q_s(x,y)^2+P_s(x,y)+7\left(\frac{107}{100}\right)^{-s}-\frac{7}{100}T_s(x,y)\right)}}, \\
U_s(x,y) &= e^{-e^{-\frac{100}{100}\left(\frac{107}{100}\right)^{-s}\left(\left(\frac{3}{2}-\frac{1}{2}\cos(ss^2)\right)L_s(x,y)+\frac{7}{100}\left(\frac{1}{5}-(Q_s(x,y))^2\right)\cos^2(30Q_s(x,y))\left(\frac{1}{2}+\frac{1}{\pi}\arctan(-100L_s(x,y))\right)-\frac{3}{250}\sin^2(120Q_s(x,y))\left(\frac{1}{2}+\frac{1}{\pi}\arctan(100L_s(x,y))\right)\right.}} \\
&\quad \left.+(Q_s(x,y))^2\left(\frac{1}{10}+\frac{T_s(x,y)}{9}\right)\left(\frac{1}{2}+\frac{1}{\pi}\arctan\left(1000(|Q_s(x,y)|-\frac{21}{50})\right)\right)\right)}, \\
T_s(x,y) &= e^{-e^{40(P_s(x,y)-\frac{1}{25}+\frac{1}{100}Q_s(x,y))}}, \\
L_s(x,y) &= P_s(x,y)-\frac{1}{10}|Q_s(x,y)|^{3/5}-\frac{7}{10}\cos(5s^2)|Q_s(x,y)|+2(Q_s(x,y))^4, \\
Q_s(x,y) &= \arctan\left(\tan\left(\frac{1}{5}\left(\frac{23}{20}\right)^s\left(x-\frac{16}{25}\cos(s^2)y\right)+4\cos(5s)\right)\right), \\
P_s(x,y) &= \arctan\left(\tan\left(\frac{1}{5}\left(\frac{23}{20}\right)^s\left(y+\frac{16}{25}\cos(s^2)x\right)+4\cos(7s)\right)\right), \\
E(x,y) &= \sum_{s=1}^{40}\left(\frac{25}{26}\right)^s N_s(x,y), \\
N_s(x,y) &= \cos\Big(100^{-1}5^45^{-s}(1+\cos(10s))(-\cos(s^2)x+\sin(s^2)y+2\cos(17s)) \\
&\quad +4\cos(100^{-1}5^45^{-s}(-\cos(9s)x+\sin(9s)y))+2\cos(5s)\Big) \\
&\quad \times \cos\Big(100^{-1}5^45^{-s}(1+\cos(10s))(\cos(s^2)y+\sin(s^2)x+2\cos(15s)) \\
&\quad +4\cos(100^{-1}5^45^{-s}(-\cos(8s)x+\sin(8s)y))+2\cos(7s)\Big).
\end{aligned}$$

4. Polar Bear Walking in Front of Sunset

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{1200},\frac{601-n}{1200}\right)\right),F\left(H_1\left(\frac{m-1000}{1200},\frac{601-n}{1200}\right)\right),F\left(H_2\left(\frac{m-1000}{1200},\frac{601-n}{1200}\right)\right)\right), \quad F(x)=\left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$\begin{aligned}
H_v(x,y) &= \frac{1-C(x,y)}{200}\Big(200A_{300}(x,y)+20(7-5v^2+10v)A_{75}(x,y) \\
&\quad + (20+3v^2-15v)(1-A_{300}(x,y))(11-4\sqrt{x^2+y^2})Z_{30}(x,y)+200R_1(x,y)\Big) \\
&\quad + \frac{20+3v^2-15v}{80}C(x,y),
\end{aligned}$$

$$C(x,y)=1-e^{-e^{12(B(x,y)+W(x,y))-e^{-40\left[\frac{1}{250}\cos(7x+80y)^2+\frac{y}{3^?}+\frac{E(x,y)}{30}\right]^2}},$$

$$A_v(x,y)=e^{-e^{\left(\frac{19}{20}x^2-y^2+\frac{1}{20}\cos(19x+17y+\cos(28x-yy))-\frac{1}{250}\right)}},$$

$$B(x,y)=-\frac{1}{\pi}\arctan\bigg(500\Big[\frac{7}{20}\left(10\left(x-\frac{1}{500}\right)-\frac{63}{100}\right)^2+7\Big|10\left(y+\frac{8}{25}\right)-\frac{1}{10}\left|\frac{1}{2}e^{\frac{5}{2}(x-\frac{1}{500})^2}\right|\right|\\ +\frac{8}{25}\sin^8\left(10x-\frac{10}{50}+\frac{1}{5}\right)-\frac{1}{10}\sin^{100}\left(10\left(x-\frac{1}{500}\right)+\frac{1}{5}\right)-\frac{28}{25}+\frac{E(x,y)}{30}\Big]\bigg)\\ -\frac{17}{20}-e^{500(10(x-\frac{1}{500})-4(y+\frac{8}{25})-\frac{43}{20})}+\frac{1}{2},$$

$$W(x,y)=\sum_{s=1}^410e^{-e^{-125s\left(10(x-\frac{8}{25})\right)^2+\frac{E(x,y)}{30}-300\left(\frac{1}{10}+100\left|y-\frac{8}{25}\right|\right)\left(10x-\frac{s}{2}+\frac{1}{10}+\frac{2}{10}+\frac{1}{10}\left(\frac{13}{10}+\frac{23}{10}-s\right)^2\right)}}\\ +\frac{1}{10}\cos^2\left(10\left(y+\frac{8}{25}\right)+\frac{1}{10}\right),$$

$$R_v(x,y)=\frac{5}{4}\sum_{s=1}^{30}Z_{s-1}(x,y)J_{50,s}(x,y)\frac{3v^2-15v+20}{20}\left(\frac{5-(v-1)^2}{2}-\frac{9-2(v-1)^2}{4}J_{20,s}(x,y)\right)\left(\frac{3}{2}-x^2-y^2\right),$$

$$Z_s(x,y)=\prod_{u=0}^s(1-J_{50,u}(x,y)),\qquad E(x,y)=\sum_{s=1}^{50}\frac{25^s}{26^s}T_s(x,y),$$

$$T_s(x,y)=\cos\Big(5^{-1}\mathfrak{I}^45^{-s}(1+\cos(10s))(\sin(s^2)y-\cos(s^2)x+2\cos(17s))\\ +4\cos(10^{-1}\mathfrak{I}^45^{-s}(\sin(9s)y-\cos(9s)x))+2\cos(5s)\Big)\\ \times\cos\Big(5^{-1}\mathfrak{I}^45^{-s}(1+\cos(10s))(\cos(s^2)y+\sin(s^2)x+2\cos(15s))\\ +4\cos(10^{-1}\mathfrak{I}^45^{-s}(\sin(8s)y-\cos(8s)x))+2\cos(7s)\Big).$$

5. Jellyfish

$$\text{rgb}\Big(F\Big(H_0\Big(\frac{m-1000}{600},\frac{651-n}{600}\Big)\Big),F\Big(H_1\Big(\frac{m-1000}{600},\frac{651-n}{600}\Big)\Big),F\Big(H_2\Big(\frac{m-1000}{600},\frac{651-n}{600}\Big)\Big)\Big),\quad F(x)=\Big[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\Big]$$

$$H_v(x,y)=\frac{19-21v+7v^2}{10}\bigg(R_0(x,y)+R_1(x,y)(1-R_0(x,y))\prod_{s=1}^{35}\bigg(1-\frac{3}{4}j_s(x,y)\bigg)\\ +\frac{9v^2}{40}\left(1+\frac{2y}{3}-\frac{x^2}{20}+\frac{E(x,y)}{20}\right)+\frac{5-3v+v^2}{5}C(x,y)(1-R_0(x,y))\left(1-\frac{3}{5}W_{500}(x,y)\right)\\ +\frac{5-3v+v^2}{5}\left(W_5(x,y)+\frac{y}{4}-\frac{3}{20}+\left(\frac{3}{10}-\frac{v}{20}+\frac{2-v}{5}\left(y-\frac{2}{5}\right)\right)K(x,y)+\frac{8-7v}{20}e^{-e^{10y-5+8(x,y)}}\right)W_{500}(x,y)U(x,y)\bigg),$$

$$U(x,y)=\frac{9}{20}+2e^{-e^{\cdots}}+\frac{3}{5}e^{-e^{\cdots}}+\frac{1}{5}e^{-e^{\cdots}},$$

$$\begin{aligned}
K(x,y) &= e^{-e^{10(\cdots)}},\\
C(x,y) &= \sum_{s=1}^{32}\left(\prod_{u=0}^{s-1}\left(1-\frac{3}{4}e^{-e^{10(\cdots)}}\right)\right)j_s(x,y)\left(e^{-e^{\cdots}}+\frac{2}{5}e^{-e^{\cdots}}+\frac{2}{5}e^{-e^{\cdots}}\right),\\
P_s(x,y) &= Q_s(x,y)+\sum_{u=0}^7\frac{50-s}{1000}\left(\frac{3}{5}\right)^u\arctan\left(3\cos\left(2^u(1+s)\arctan\left(\frac{1000x}{250(2y-1)^2+1}\right)+\frac{1}{2}\left(\frac{7}{5}\right)^u\cos(2^u(5+s)Q_s(x,y))+2us\right)\right),\\
E(x,y) &= \sum_{s=1}^{50}\frac{25^s}{26^s}\frac{T_s(x,y)}{10},\\
R_p(x,y) &= e^{-e^{-50\left(\sin\left(v+30\arccos\left(\frac{9}{5}L_p(x,y)\left(\frac{1}{2}+\frac{1}{\pi}\arctan(-100)L_p(x,y)\right)\right)\right)+\frac{1}{50}-\frac{y^6}{10}+\frac{2}{\pi}\arctan(\cdots)}\right)},\\
Q_s(x,y) &= y-\frac{10-s}{30}+\frac{1}{5}-x^2+\frac{\cos(8x+5s)}{10},\\
T_s(x,y) &= \cos\big(23^s20^{-s}10(1+\cos(10s))(\cos(7s)x+\sin(7s)y+2\cos(17s))+4\cos\big(23^s20^{-s}10(\cos(7s)x+\sin(7s)y)\big)+2\cos(5s)\big)\\
&\quad\times\cos\big(23^520^{-5}10(1+\cos(10s))(\cos(7s)y-\sin(7s)x+2\cos(15s))+4\cos\big(23^520^{-5}10(\cos(8s)x+\sin(8s)y)\big)+2\cos(7s)\big).
\end{aligned}$$

6. Whirlpool

$$\text{rgb}\Big(F\Big(H_0\Big(\frac{m-1000}{500},\frac{501-n}{500}\Big)\Big),F\Big(H_1\Big(\frac{m-1000}{500},\frac{501-n}{500}\Big)\Big),F\Big(H_2\Big(\frac{m-1000}{500},\frac{501-n}{500}\Big)\Big)\Big),\quad F(x)=\Big[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\Big]$$

$$\mathsf{P}_v(y)=y+\frac{v}{200}.$$

7. Active Volcano

$$\text{rgb}\Big(F\Big(H_0\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big),F\Big(H_1\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big),F\Big(H_2\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big)\Big),\quad F(x)=\Big[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\Big]$$

$$\begin{aligned}
H_v(x,y) &= \frac{2v^2-v+12}{20}(1-W_1(x,y))\left(\frac{11}{10}-\frac{x}{25}+\frac{y}{40}\right)\left(1+\frac{3v^2-13v+2}{20}e^{-e^{3y+\frac{9}{2}}}\right)(1-R(x,y))\\
&\quad +W_1(x,y)\left(\frac{v^2-v+2}{40}+\frac{34-9v+3v^2}{40}e^{-e^{1000(T_1(x,y)-T_0(x,y))}}\right)(1-R(x,y))\\
&\quad + (250+175(v^2-v))\,e^{-e^{30|\cdots|}}\left(\frac{33}{250}\frac{3}{100}\left(y-\frac{7}{20}\right)^{10}\right)\\
&\quad +\frac{9+3B_1(x,y)}{20}\left(\frac{12-v}{50}+\frac{3v^2-8v+11}{10}e^{-e^{-500(C_1-C_0)}}\right)R(x,y),
\end{aligned}$$

$$R(x,y)=e^{-e^{-50\left(\left(\frac{7}{3}+P_1(y)\right)^4+\frac{1}{2}(P_1(y))^2-\left(Q_1(x)+\frac{P_1(x)}{2}\right)^2-A_1(x,y)\frac{B_1(x,y)}{45}\right)}}e^{-e^{-1000\left(x+\frac{4}{3}-\frac{1}{2}\right)}},$$

$$T_v(x,y)=\sum_{s=1}^{50}\frac{17^s}{20^s}e^{-e^{s-10\cos(\cdots)}}\cos\bigg(\frac{1}{20}3^s(\cos(3s^2)Q_vM_v+2\sin(3s^2)S_v+2\cos(5s^2))\bigg)\\ \times\cos\bigg(\frac{1}{20}6^s(\sin(3s^2)Q_vM_v-2\cos(3s^2)S_v+2\cos(7s^2))\bigg),$$

$$A_v(x,y)=\sum_{s=1}^{50}\left(\frac{87}{100}\right)^s\left(K_{v,s}(x,y)L_{v,s}(x,y)\right)^4,$$

$$C_v(x,y)=\sqrt{\left|5\left[\frac{7}{5}+P_v(y)+\frac{P_v(y)^4}{2}-\left(Q_v(x)+\frac{P_v(y)^2}{2}\right)-A_v(x,y)\right]\right|},$$

$$B_v(x,y)=\sum_{s=1}^{50}2\left(\frac{197}{200}\right)^s\left(\frac{3}{10}-(K_{v,s}(x,y))^6\right)\left(\frac{3}{10}-(L_{v,s}(x,y))^6\right),$$

$$S_v(x,y)=\sqrt{(V_v(x,y))^2+\frac{1}{3}(Q_v(x))^2},$$

$$K_{v,s}(x,y)=\cos\Big(2^{-1}23^s20^{-s}(\cos(s^2)Q_v(x)+2^{-1}3\sin(s^2)P_v(y))\\ +2^{-1}\cos(2^{-1}23^s20^{-s}3(\cos(2s^2)Q_v(x)+2^{-1}3\sin(2s^2)P_v(y))+\cos(7s^2))\Big),$$

$$L_{v,s}(x,y)=\cos\Big(2^{-1}23^s20^{-s}(2^{-1}3\cos(s^2)P_v(y)-\sin(s^2)Q_v(x))\\ +2^{-1}\cos(2^{-1}23^s20^{-s}3(\cos(3s^2)Q_v(x)+2^{-1}3\sin(3s^2)P_v(y))+\cos(8s^2))\Big),$$

$$W_v(x,y)=e^{-e^{1000V_v(x,y)-1000\Big(7+800|200Q_v(x)-\frac{7}{2}|^2\Big)^{-3/2}-8\cos^2(32P_v(y)-19Q_v(x)-\cos(12P_v(y)+7Q_v(x)))-4\cos^2(82P_v(y)+49Q_v(x)-2\cos(32P_v(y)-27Q_v(x)))+400+\frac{1000}{23}|Q_v(x)|^5}},$$

$$M_v(x,y)=50^{5/2}(23|5V_v(x,y)+2|)^{5/2},$$

$$V_v(x,y)=P_v(y)-20^{-1}\cos^4(4Q_v(x)+3P_v(y)+\cos(2Q_v(x)+P_v(y)))-500^{-1}9\cos^2(14P_v(y)-8Q_v(x)+\cos(5P_v(y)-3Q_v(x))),$$

$$P_v(y)=y+10^{-3}v,\qquad Q_v(x)=x+500^{-1}v.$$

8. Strawberries

$$\mathrm{rgb}\Big(F\Big(H_0\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big),F\Big(H_1\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big),F\Big(H_2\Big(\frac{m-1000}{600},\frac{601-n}{600}\Big)\Big)\Big),\quad F(x)=\Big[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\Big]$$

$$H_v(x,y)=\sum_{s=1}^{30}\left(\prod_{r=0}^{s-1}(1-A_{1000,r}(x,y))\left(1-e^{-e^{-1000(r-s)}}\right)U_s(x,y)\left(1-\frac{5}{4}A_{4,r}(x,y)\right)\right)\\ \times\left(\frac{5-3(v-1)^2+W(x,y)}{10}e^{-e^{-\frac{24}{10}10P_s(x,y)}}U_s(x,y)+A_{1000,s}(x,y)(1-U_s(x,y))L_{v,s}(x,y)\right),$$

$$L_{v,s}(x,y)=\left(\frac{1}{10}-\frac{1}{40}\cos(20\arccos(R_{0,s}(x,y)))\cos(25P_s(x,y))\right)$$

$$\begin{aligned}
& \times \left(4v^2 - 13v + 11 + \cos(7s + vs) + 20e^{-e^{-70(r_{0,s}(x,y) - \frac{1}{2})}} + 20e^{-e^{-10(r_{0,s}(x,y) - \frac{1}{2})}} \right) A_{4,s}(x,y) A_{1000,s}(x,y) + B_s(x,y), \\
C_v(x,y) &= e^{-e^{-\left(\cos(10 \arccos(R_{0,s}(x,y))) \cos(\frac{25}{3} P_s(x,y) - \frac{W(x,y)}{10} - \frac{2}{3}) - e^{-(\cos(10 \arccos(R_{0,s}(x,y))) \sin(\frac{25}{3} P_s(x,y) - \frac{W(x,y)}{10} - \frac{2}{3}))} \right)}}, \\
B_s(x,y) &= e^{-e^{-70(\cos(2s \arccos(R_{0,s})) \cos(2s P_s) - \frac{48}{40})}}, \\
A_{v,s}(x,y) &= e^{-e^{-1000(s - \frac{1}{2}) - e^{-\frac{10}{50}(1-s) \left((R_{0,s}(x,y))^2 + (P_s(x,y))^2 \right) - \frac{11}{20} \arctan(1000(x-100))}}}, \\
U_s(x,y) &= 1 - (1 - M_s(x,y))(1 - N_s(x,y)), \\
M_s(x,y) &= e^{-e^{-10 \left(P_s(x,y) - \frac{s^2}{100} + \frac{s}{20} \right) \left(\frac{s-1}{10} \cos(7s(x,y)+25) - \cos((10+3 \cos(14s)) \arccos(R_{0,s}(x,y))) + \frac{s}{100} \cos(45x+7y - \cos(15x) + 2 \cos(5s)) \right)}}, \\
N_s(x,y) &= e^{-e^{-100 \left(P_s(x,y) - \frac{37}{50} \right) \left(\frac{5}{20} \frac{1}{10} \cos(\alpha_s) + \frac{1}{10} \cos(\beta_s) \right)}}, \\
R_{t,s}(x,y) &= E_{t,s}(x,y) e^{-e^{-c_{t,s}(x,y)}}, \\
E_{t,s}(x,y) &= \frac{1000}{\sqrt{20}} Q_s(x,y) \sqrt{5 [20 - 20(1-2t)P_s(x,y) - 27t] \left(1 + 50 \sqrt{14(200 - (20(1-2t)P_s(x,y) + 27t)^2)} \right)^{-1}}, \\
P_s(x,y) &= \arctan \left(\tan \left(2 \sin(5s)x - 2 \cos(5s)y + 3 \cos(5s) + \frac{3 \cos(14x - 19y + 5s)}{200} \right) \right), \\
Q_s(x,y) &= \arctan \left(\tan \left(2(\cos(5s)x + \sin(5s)y + 2 \cos(4s)) + \frac{3 \cos(18x + 15y + 4s)}{200} \right) \right), \\
W(x,y) &= \sum_{s=1}^{40} e^{-e^{-y_j \cos^2(2u_u s^{-1} \cos(\dots)) \cos^2(2v_u s^{-1} \cos(\dots)) - \frac{97}{100}}}.
\end{aligned}$$

9. Bird Flying Over a Leafless Forest

$$\text{rgb} \left(F \left(H_0 \left(\frac{m-1000}{600}, \frac{601-n}{600} \right) \right), F \left(H_1 \left(\frac{m-1000}{600}, \frac{601-n}{600} \right) \right), F \left(H_2 \left(\frac{m-1000}{600}, \frac{601-n}{600} \right) \right) \right), \quad F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$$

$$H_v(x,y) = \frac{1+4A(x,y)}{250} (7+3K(x,y))(10-7R(x,y)) \left(\frac{43}{100} - \frac{v}{8} + \frac{v^2-v}{25} \right),$$

$$K(x,y) = \sum_{s=1}^{15} \frac{5+\cos(7s)}{5} J_s(x,y) \prod_{u=0}^{s-1} \left(1 - \frac{1}{2} J_u(x,y) \right),$$

$$A(x,y) = \prod_{s=1}^{15} Z_s(x,y), \quad E(x,y) = \sum_{s=1}^{50} \frac{25^s}{26^s} O_s(x,y),$$

$$J_s(x,y) = e^{-e^{-100(s-\frac{1}{2})-\Phi_s(x,y)}},$$

$$Z_s(x,y) = W_{0,s}(x,y) M_{0,s}(x,y) G_s(x,y),$$

$$G_s(x,y) = e^{-e^{-15(\text{angular terms})+\text{phase}}},$$

$$\begin{aligned}
M_{0,s}(x,y) &= e^{-e^{\Psi_{0,s}(x,y)}}, & W_{0,s}(x,y) &= e^{-e^{\Psi_{1,s}(x,y)}}, \\
G_v(x,y) &= e^{-e^{\Psi_2(x,y)}}, \\
N_s(x,y) &= \cos(D_s(x,y)), \\
D_s(x,y) &= 20\sqrt{x^2+y^2} + 4\arccos(B_s(x,y)) + 2^{-1}\cos\left(9\left(\sqrt{x^2+y^2} + 5^{-1}\arccos(B_s(x,y)) + 4s + S_s(x,y)\right)\right) \\
&\quad + x^2 + 10L_s(x,y), \\
R_s(x,y) &= \cos((5+2x)P(x,y)) + \cos((2+s)P(x,y)) + 49^{-1}(5+2s)P_s(x,y) + 3\cos(s^2), \\
L_s(x,y) &= 10^{-1}\cos(4Q_s + 2P_s + \cos(2Q_s - R_s + s) + 4s) + 20^{-1}\cos(100Q_s - 7R_s + \cos(4Q_s + 3R_s - s) + 3s) \\
&\quad + 30^{-1}\cos(23Q_s + 10P_s + \cos(14Q_s - 6R_s + 2s) + 5s) + 100^{-1}\cos(50Q_s - 18P_s + \cos(24Q_s - 16P_s + s) + 7s), \\
P(x,y) &= \arctan\left(-\frac{10000y}{10000x+1}\right) + e^{-e^{\cdots}}\pi, \\
R(x,y) &= e^{-e^{\cdots}}, \\
V(x,y) &= (y-x-3(U(x,y))^4 + 7T(x,y))\frac{70^{-1}-10(U(x,y))^2}{10}, \\
T(x,y) &= e^{-e^{1004(y+\cdots)}}, & U(x,y) &= \frac{5}{4}(10^{-4}-x-y), \\
Q_1(x,y) &= (x^2+y^2+10^{-4})^{1/2}, \\
O_s(x,y) &= \cos\left(5^s4^{-s}10(1+\cos(10s))(\sin(s^2)y - \cos(s^2)x + 2\cos(17s)) + 4\cos(5^s4^{-s}5(\sin(9s)y - \cos(9s)x)) + 2\cos(5s)\right) \\
&\quad \times \cos\left(5^s4^{-s}10(1+\cos(10s))(\cos(s^2)y + \sin(s^2)x + 2\cos(15s)) + 4\cos(5^s4^{-s}5(\sin(8s)y - \cos(8s)x)) + 2\cos(7s)\right).
\end{aligned}$$

∞

10. Interacting Stars

$$\begin{aligned}
&\text{rgb}\left(F\left(H_0\left(\frac{m-1040}{480}, \frac{601-n}{480}\right)\right), F\left(H_1\left(\frac{m-1040}{480}, \frac{601-n}{480}\right)\right), F\left(H_2\left(\frac{m-1040}{480}, \frac{601-n}{480}\right)\right)\right), & F(x) &= \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right] \\
H_v(x,y) &= \left(\frac{6-9v+3v^2}{8} + \frac{2-3v^2+5v}{8}C_0(x,y)\right)e^{-e^{10((U_0(x,y))^2+(V_0(x,y))^2)-4}} \\
&\quad + \frac{7}{5}\left(\frac{v^2-4v+4}{4} + \frac{C_1(x,y)}{2}\right)e^{-e^{10((U_1(x,y))^2+(V_1(x,y))^2)-\frac{4}{25}}} \\
&\quad + \left(3 - \frac{3v}{2} + \frac{3(v^2-v)}{8}\right)\left(L(x,y) + \frac{K_0(x,y)}{10} + K_1(x,y)\frac{5-(v-1)^2}{20}\right), \\
C_v(x,y) &= \sum_{s=1}^{50} \frac{3}{10} \left(\frac{24}{25}\right)^s e^{-e^{-10s}\{\text{oscillatory terms in } x,y,s\}}, \\
P_{v,s}(x,y) &= \arcsin\left(\frac{W_v(x,y)}{2}(\cos(s^2)V_v(x,y) - \sin(s^2)U_v(x,y))\right), \\
Q_{v,s}(x,y) &= \arcsin\left(\frac{W_v(x,y)}{2}(\cos(s^2)U_v(x,y) + \sin(s^2)V_v(x,y))\right),
\end{aligned}$$

$$\begin{aligned}
K_v(x,y) &= \sum_{s=1}^{50} e^{-e^{\Omega_{v,s}(x,y)}}, \quad W_v(x,y) = e^{-e^{\omega_v(x,y)}}, \\
L(x,y) &= \sum_{s=1}^{50} \frac{1}{10} e^{-e^{\Lambda_s(x,y)}}, \\
V_v(x,y) &= \left(y - \frac{1}{2} + \frac{7v}{10} + 10^{-4}\right) T(x,y), \\
U_v(x,y) &= \left(x - \frac{21}{10} + \frac{7v}{2} + 10^{-4}\right) T(x,y), \\
R(x,y) &= \left(1 + \frac{1}{2} \left(x + \frac{y}{6} + \frac{9}{20}\right)\right) \cos(2s^2) \left(x + \frac{y}{6} + \frac{9}{20}\right)^2, \\
T(x,y) &= 1 - \frac{1}{5} e^{-2(x^2+y^2)-8(y^2)}.
\end{aligned}$$

11. Supernova Remnant

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{540}, \frac{601-n}{540}\right)\right), F\left(H_1\left(\frac{m-1000}{540}, \frac{601-n}{540}\right)\right), F\left(H_2\left(\frac{m-1000}{540}, \frac{601-n}{540}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$H_v(x,y)=A_v(x,y)+e^{-e^{-10[(x-y)^2+(y-2)^2]}}G_v(x,y),$$

$$G_v(x,y)=\sum_{s=1}^{50}\left(\frac{5}{2}+\frac{5}{2}(v-1)^2+(3-3v)(-1)^v\right)e^{-e^{-2000\cos(\cdots)}},$$

$$\begin{aligned}
A_v(x,y) &= \sum_{s=1}^{50} \left(\prod_{u=0}^{s-1} \left(1 - \frac{3}{5} J_{v,u}(x,y) \right) \right) J_{v,s}(x,y) \left(\frac{1}{4} + J_{0,s}(x,y) (1 + 6L(x,y)) \right) \\
&\quad \times \left(1 - \frac{17v}{20} + \frac{19}{40} (v^2 - v) + \frac{v-1}{2} \cos(5s^2) + \frac{(2-v)^2 - 2 + v}{5} \cos(9s^2) \right),
\end{aligned}$$

$$J_{v,s}(x,y)=e^{-e^{-10(s-\frac{1}{2})-(\frac{3-3v}{20})^2}U_{v,s}(x,y)\cos^2(\cdots)-e^{\cdots}},$$

$$V_s(x,y)=\arctan\!\!\left(\frac{\cos(10s)Q_s(x,y)+\sin(10s)P_s(x,y)}{\cos(10s)P_s(x,y)-\sin(10s)Q_s(x,y)}\right),$$

$$U_s(x,y)=Q_s(x,y)^2+P_s(x,y)^2,$$

$$P_s(x,y)=\frac{3}{2}\left(y-\frac{x}{2}\right)+\frac{1}{5}\cos(-3y+7s)+\frac{1}{20}\cos(26y+6s)+\frac{1}{100}\cos(-21y+2s)+\frac{1}{250}\cos(59y+23s),$$

$$Q_s(x,y)=x+\frac{y}{2}+\frac{1}{10}\cos(19s)+\frac{1}{5}\cos(5x+2y+7s)+\frac{1}{20}\cos(15x-4y+9s)+\frac{1}{100}\cos(35x+63y+12s)+\frac{1}{250}\cos(135x+120y+17s),$$

$$I(x,y)=\sum_{s=1}^{50}e^{-e^{-20\cos(\cdots)\cos(\cdots)-47/50}}.$$

12. Dust Storm

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_1\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_2\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$\begin{aligned} H_v(x, y) &= \frac{5-v}{5} \left(1 - |M(x, y) - 1|e^{-1000(M(x, y)-1)}\right) \\ &\quad + \frac{v^2 - 5v + 14}{20} \left(1 - \frac{y}{10}(1 - R_1(x, y))R_0(x, y)\right) \left(\frac{6 - 3v^2 + 5v}{20} + \frac{P(x, y)}{50} + \frac{3v^2 - 5v + 14}{20}R_0(x, y)\right) Z_{30}(x, y)e^{-1000v} \\ &\quad + \left(7v^2 - 7v + 186 + 16e^{-e^{\Xi(x, y)}}\right) \frac{Z_{30}(x, y)}{200}e^{-10(v, m)}, \end{aligned}$$

$$A(x, y) = \sum_{s=1}^{50} Z_{s-1}(x, y) J_{100, s}(x, y) \left(2 - \frac{J_{5s}(x, y)}{s} - \frac{29}{10} J_{2s}(x, y)\right) \frac{3 + 2C_{s-1}(x, y)}{5} \left(\frac{y}{5} + \frac{9}{10} - \frac{x^2}{50000}\right),$$

$$R_0(x, y) = \prod_{s=1}^{20} e^{-e^{\Phi_s(x, y)}},$$

$$W(x, y) = \frac{200}{200 - 3v^2} + \frac{300}{300y + 51 + P(x, y)}, \quad Z_s(x, y) = \prod_{u=0}^s \left(1 - \frac{J_{100, u}(x, y)}{2}\right),$$

$$J_{5s}(x, y) = e^{-e^{\Psi(x, y)}}, \quad C_s(x, y) = e^{-e^{\Omega(x, y)}},$$

$$K_v(x, y) = e^{-e^{|\Theta_v(x, y)|}},$$

$$L_1(x, y) = (M_1(x, y))^2 + 3(W_1(x, y))^2,$$

$$N_s(x, y) = \frac{5y}{9} + \frac{1}{5} - \frac{3}{10} \left(6 - s\sqrt{\frac{5}{40}}\right) \cos^2(4s) + \frac{3}{250}P(x, y) + \frac{\cos(10s)}{40},$$

$$M_s(x, y) = \frac{5x}{9} - \frac{94}{40} - \frac{3}{10} \cos^2(4s) + \frac{5}{18} + \frac{3}{250}Q(x, y),$$

$$P(x, y) = \sum_{s=1}^{50} \frac{19^s}{20^s} \cos\left(\frac{5^s}{\pi} U_s(x, y) + 5 \cos(Q_s(x, y))\right),$$

$$Q(x, y) = \sum_{s=1}^{50} \frac{19^s}{20^s} \cos\left(\frac{5^s}{\pi} U_s(x, y) + 5 \cos(Q_s(x, y))\right),$$

$$U(x, y) = \frac{5^s}{4 \cdot 10} (\cos(27s)V_s(x, y) - \sin(27s)U_s(x, y)) + 2 \cos(10s),$$

$$G_0(x, y) = 10^{-1}5^45^{-s} (\cos(23s)U_s(x, y) + \sin(23s)V_s(x, y)) + 2 \cos(10s),$$

$$V_0(x, y) = \cos(16s)x + \sin(16s)y + 2 \cos(10s),$$

$$V(x, y) = \cos(16s)y - \sin(16s)x + 2 \cos(14s).$$

13. Eye of a Storm

$$\text{rgb}\left(F\left(H_0\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_1\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right), F\left(H_2\left(\frac{m-1000}{600}, \frac{601-n}{600}\right)\right)\right), \quad F(x) = \left[255e^{-e^{-1000x}}|x|e^{-e^{1000(x-1)}}\right]$$

$$H_v(x, y) = A_v(x, y)W(x, y) + \frac{2-v}{20}(1 - W(x, y)),$$

$$W(x, y) = e^{-e^{-1000\left[\left(\frac{y}{100} - \frac{x}{100}\right)^2 + y^2\right]^{1/2}}} e^{-e^{-100\left[y - \frac{x}{10} + \frac{1}{10}(2xy-3)^2\right]}} e^{-e^{-100\left[\frac{y}{100} - \frac{1}{100}\right](2xy-10)}},$$

$$A_v(x, y) = \sum_{s=1}^{50} \left(\prod_{r=0}^{s-1} (1 - J_r(x, y)) \right) J_s(x, y) \left(\frac{11}{20} + \frac{3v^2 - v}{40} + \frac{17}{20}s^{-1} \right) \\ \times e^{-e^{(x, y)}} - e^{-e^{(x, y)}} - \frac{1}{100} + \frac{1}{2}e^{-e^{(x, y)}} + \frac{v}{100}\cos^2(12s),$$

$$J_s(x, y) = e^{-e^{-100(s-\frac{1}{2})}} e^{-e^{-100\Psi_s(x, y)}} e^{-e^{-100\Omega_s(x, y)}},$$

$$K_s(x, y) = \cos\left(\frac{6}{5}(3 + 2\cos(19s))U_{v,s}(x, y) + 4\sin(8s)\right) \\ \times \cos(2(3 + 2\cos(19s))V_{v,s}(x, y) + 4\sin(17s)),$$

$$V_{v,s}(x, y) = P_{v,s}(x, y) + \frac{2}{5}\cos\left(\frac{2}{5}P_{v,s}(x, y) - \frac{3}{10}Q_{v,s}(x, y) + 8s\right) + \frac{3}{10}\cos(2Q_{v,s}(x, y) - P_{v,s}(x, y) + 3s) \\ + \frac{1}{20}\cos(4Q_{v,s}(x, y) + 4P_{v,s}(x, y) + 3s),$$

$$U_{v,s}(x, y) = Q_{v,s}(x, y) + \frac{2}{5}\cos\left(\frac{1}{2}P_{v,s}(x, y) + \frac{3}{5}Q_{v,s}(x, y) + 6s\right) + \frac{3}{10}\cos(P_{v,s}(x, y) + 2Q_{v,s}(x, y) + 2s) \\ + \frac{1}{20}\cos(5P_{v,s}(x, y) - 3Q_{v,s}(x, y) + 2s),$$

$$E(x, y) = \sum_{s=1}^{50} \frac{25^s}{26^s} C_s(x, y),$$

$$P_{v,s}(x, y) = \arctan\left(\frac{\cos(2\ln(Q_{v,s}(x, y)))(x + \frac{3y}{2} + \frac{s}{50}) + \frac{6}{5}\sin(2\ln(Q_{v,s}(x, y)))(\frac{3y}{2} + \frac{s}{50} - x - \frac{v}{100})}{10^{-6} + \frac{2}{5}\cos(2\ln(Q_{v,s}(x, y)))(\frac{3y}{2} + \frac{s}{50} - x - \frac{v}{100}) - \sin(2\ln(Q_{v,s}(x, y)))(x + \frac{3y}{2} + \frac{s}{50})}\right),$$

$$Q_{v,s}(x, y) = \sqrt{\left(x + \frac{3y}{2} + \frac{s}{50}\right)^2 + \left(\frac{3y}{2} + \frac{s}{50} - x - \frac{v}{100}\right)^2 + 10^{-6}},$$

$$C_s(x, y) = \cos\Big(6^s5^{-s}10(1+\cos(10s))(\cos(7s)x+\sin(7s)y+2\cos(17s)) \\ +4\cos\big(6^s5^{-s}10(\cos(9s)x+\sin(9s)y)+2\cos(5s)\big) \\ \times\cos\Big(6^s5^{-s}10(1+\cos(10s))(\cos(7s)y-\sin(7s)x+2\cos(15s)) \\ +4\cos\big(6^s5^{-s}10(\cos(8s)x+\sin(8s)y)+2\cos(7s)\big)\Big).$$